Abdullah Al Noman

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Education

Ph.D. in Computer Science 2025 – Present
University of Texas at Arlington, USA *CGPA: 3.71/4.0*
Transformative Wireless Systems and Technology (TWiST) Lab
PhD Supervisor: **Dr. Debashri Roy**

Research Focus: Space–Air–Ground Integrated Networks (SAGIN), Direct-to-Satellite (D2S), Wireless PHY/MAC Optimization, AI/ML-based Resource Management, Non-Terrestrial Networks (NTN), O-RAN (Open Radio Access Network), Next-Generation (NextG) Wireless Networks

Bachelor of Technology (B.Tech.) in Computer Science & Engineering 2020 – 2024
National Institute of Technology Tiruchirappalli, India *CGPA: 7.51 / 10.00 (75.1%)*

Publications

- **A. A. Noman**, G. Singh, R. Ahmed, and D. Roy, "IRIS: Intelligent Random Access with Integrated Shared Learning for Decentralized LEO Direct-to-Satellite Uplinks," *arXiv preprint*, 2026. [arXiv]
- **A. A. Noman**, S. S. Reza, M. Yuksel, and D. Roy, "From Orbit to Device: Spectrum Intelligence for Multi-Tier LEO-Aided D2C IoT Networks," *IEEE Internet of Things Magazine*, 2026, **Accepted**.
- M. Kulhandjian, **A. A. Noman**, and D. Roy, "Graph-Spectral Identifiability and Cramér–Rao Bounds for Multistatic TDOA Geolocation with Uncertain Sensor Poses," *IEEE Military Communications Conference (MILCOM)*, 2026, **Accepted**.
- T. T. Sari, R. Ahmed, **A. A. Noman**, G. Seçinti, C. Dick, and D. Roy, "TITAN: Twin-Informed Topology Adaptation for LAWN-enabled D2C Communication," *IEEE Transactions on Vehicular Technology (TVT)*, 2026, **Accepted**. [IEEE Xplore]
- **A. A. Noman**, T. T. Sari, S. Amatare, G. Seçinti, and D. Roy, "Space-Air-Ground Network for Direct-to-Cell Communication," *IEEE Military Communications Conference (MILCOM)*, 2025. [IEEE Xplore]
- M. Priyadarsini, **A. A. Noman**, V. Patel, and S. Prakash, "A Reliable and Secure Multicast Transport Protocol for Space-Ground Integrated Networks using RBAC," *15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, IEEE, 2024. [DOI]

Experience

Graduate Research Assistant (GRA)
University of Texas at Arlington – TWiST Lab *Arlington, TX*

- Conducted research on next-generation 5G/6G wireless systems, focusing on non-terrestrial networks (NTN), direct-to-satellite communications, and intelligent resource management.
- Developed MATLAB- and Python-based simulation frameworks to evaluate wireless network performance in terms of throughput, latency, and reliability.
- Investigated AI/ML and reinforcement learning techniques for adaptive waveform selection and resource optimization in dynamic wireless environments.

Graduate Teaching Assistant
University of Texas at Arlington *Arlington, TX*

- Assisted with undergraduate and graduate course instruction, laboratory sessions, technical discussions, and problem-solving activities.
- Evaluated assignments, examinations, projects, and laboratory reports using standardized rubrics while providing constructive feedback.

Machine Learning Intern
Indian Institute of Technology (IIT) Ropar *Punjab, India*

- Developed and implemented machine learning algorithms for **image enhancement and green screen removal**, optimizing real-time processing performance.
- Improved computational efficiency and achieved **93% accuracy** in image processing tasks.

Summer Research Intern
National Institute of Technology Tiruchirappalli *Tamil Nadu, India*

- Developed an **Intrusion Detection System (IDS)** for smart homes using Raspberry Pi and IoT-enabled sensors to identify unauthorized activities.
- Integrated sensor data with a centralized control unit to enable real-time intrusion detection, automated alerts, and notifications.

Technical Skills

Programming: Python, C++, C, Dart
Wireless Simulation & Modeling: MATLAB, NVIDIA Sionna
AI/ML Frameworks: PyTorch, TensorFlow, Keras
Wireless Platforms & Architectures: OpenAirInterface (OAI), O-RAN, USRP SDR
Signal Processing & Localization: TDOA, Spherical Interpolation
Development Tools: Git, Jupyter Notebook, VS Code, Makefile

Projects

Development of an O-RAN System | *OpenAirInterface (OAI), 5G NR, Near-RT RIC, xApps*

- Contributing to the design and implementation of an Open Radio Access Network (O-RAN) system using OpenAirInterface (OAI) 5G NR to support flexible and interoperable wireless network architectures.
- Developing and optimizing RAN Intelligent Controller (RIC) functionalities to enable intelligent radio resource management and adaptive network control.
- Working with Near-Real-Time RIC and xApp-based control mechanisms to support dynamic network optimization and real-time decision-making in 5G radio access networks.

Spherical Interpolation for Real-Time Emitter Navigation | *USRP SDR, OctoClock, MATLAB, TDOA*

- Designed and implemented a synchronized four-node Universal Software Radio Peripheral (USRP) system using OctoClock for coherent IQ signal acquisition and multi-receiver localization.
- Implemented a MATLAB-based Spherical Interpolation (SI) algorithm for emitter position estimation using time-difference-of-arrival (TDOA) measurements from synchronized SDR receivers.
- Conducted indoor static and mobile experiments to evaluate localization performance and robustness under varying emitter positions and mobility conditions.

Reviewer Roles

- IEEE/ACM Transactions on Networking(TNET)
- IEEE 103rd Vehicular Technology Conference: VTC2026-Spring
- IEEE Transactions on Mobile Computing (TMC)
- Elsevier Computer Communications
- IEEE Global Communications Conference (GLOBECOM 2025)
- The 21st International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob 2025)
- The 18th International Workshop on Selected Topics in Wireless and Mobile Computing (STWiMob 2025)
- The 34th International Conference on Computer Communications and Networks (ICCCN 2025)
- EuCNC & 6G Summit (2025)

Achievements

- Awarded the prestigious Indian Council for Cultural Relations **(ICCR)** scholarship by the Government of India to pursue a B.Tech in Computer Science & Engineering in India.
- Honored with the Islami Bank Merit Scholarship for exceptional performance in the 10th standard board exam.
- Granted a government scholarship by the People's Republic of Bangladesh for achieving above 90% marks in both the 5th and 8th grades.